LSTM-Based Multivariate Air Quality Forecasting: PM2.5 Prediction

1. Abstract

This report presents an end-to-end implementation of a multivariate Long Short-Term Memory (LSTM) neural network for hourly PM2.5 air pollution forecasting. Using five years of meteorological and pollution data recorded in Beijing, China, the model ingests eight weather variables alongside historical pollution readings to predict the PM2.5 concentration one hour ahead. The trained model achieves a Root Mean Squared Error (RMSE) of 25.31 μg/m³, a Mean Absolute Error (MAE) of 14.00 μg/m³, and a coefficient of determination (R^2) of 0.9363 on the held-out test set, demonstrating strong predictive capability. The extremely high MAPE value observed is analysed and attributed to near-zero denominator pollution values rather than poor model fit.

2. Dataset Description

2.1 Source and Overview

The dataset is sourced from Kaggle and contains hourly weather and pollution measurements recorded at a monitoring station in Beijing, China, from January 2010 to December 2014 (five years, approximately 43,800 records). The data was originally published by Zhang et al. and is widely used as a benchmark for air quality forecasting research.

2.2 Feature Description

The complete list of raw features is presented in Table 1. After parsing date components into a unified datetime index and applying one-hot encoding to the categorical wind direction variable, the model receives a total of 11 numerical input features.

Feature	Description	Type	Unit
No	Row number (index)	Integer	—
year / month / day / hour	Timestamp components	Integer	—
pm2.5	PM2.5 concentration (target)	Float	μg/m³
DEWP	Dew point temperature	Float	°C
TEMP	Ambient temperature	Float	°C
PRES	Atmospheric pressure	Float	hPa
cbwd	Combined wind direction (categorical)	String	—
Iws	Cumulated wind speed	Float	m/s
Is	Cumulated hours of snow	Integer	hours
Ir	Cumulated hours of rain	Integer	hours

3. Methodology

3.1 Data Preprocessing

The preprocessing pipeline performs the following steps in order. (1) Datetime parsing: year,

month, day, and hour columns are combined into a single pandas DatetimeIndex to enable temporal operations. (2) Column renaming: column names are normalised (e.g., 'pm2.5' → 'pm25', 'DEWP' → 'dewp') to ensure compatibility across training and test files. (3) Wind direction encoding: the categorical variable cbwd is one-hot encoded, producing four binary indicator columns (wd_NE, wd_NW, wd_SE, wd_cv) so the model can consume directional information numerically. (4) Missing value handling: rows with missing PM2.5 values are dropped; remaining gaps are forward-filled then backward-filled to preserve temporal continuity without introducing look-ahead bias. (5) Feature alignment: only columns common to both the training and test DataFrames are retained, guaranteeing consistent input dimensionality. (6) Min-Max normalisation: all features are scaled to [0, 1] using a MinMaxScaler fitted exclusively on the training data and applied to the test data, preventing data leakage.

3.2 Sliding-Window Sequence Construction

A supervised learning dataset is constructed from the normalised time series using a sliding window of N_STEPS = 24 (hours). Each sample $X_i \in \mathbb{R}^{24\times F}$ represents the last 24 hourly observations across all F features, and the corresponding label y_i is the normalised PM2.5 value at the next time step. This formulation allows the LSTM to capture a full diurnal cycle of atmospheric variability.

3.3 Model Architecture

The LSTM model is built using the Keras Sequential API with six layers. Layer 1 is an LSTM with 128 units (return_sequences=True), which processes the full 24-step input sequence and outputs a hidden-state sequence at every time step. Layer 2 is a Dropout layer (rate=0.2) that randomly zeroes 20% of units during training to reduce overfitting. Layer 3 is a second LSTM with 64 units (return_sequences=False), compressing the sequence into a single fixed-length context vector. Layer 4 is another Dropout layer (rate=0.2) for additional regularisation. Layer 5 is a Dense layer with 32 units and ReLU activation, acting as a non-linear projection. Layer 6 is the output Dense layer with a single linear unit, predicting the next-hour normalised PM2.5 value. The model is compiled with the Adam optimiser (initial learning rate 1×10^{-3}) and MSE loss.

Layer	Type	Output Shape	Parameters
1	LSTM (128 units)	(batch, 24, 128)	72,192
2	Dropout (0.2)	(batch, 24, 128)	0
3	LSTM (64 units)	(batch, 64)	49,408
4	Dropout (0.2)	(batch, 64)	0
5	Dense (32, ReLU)	(batch, 32)	2,080
6	Dense (1, linear)	(batch, 1)	33
Total	—	—	123,713

3.4 Training Strategy

Training employs two callbacks to optimise generalisation. Early Stopping (patience=10) monitors validation loss and halts training if no improvement is observed for 10 consecutive epochs,

restoring the best weights at termination. ReduceLROnPlateau (factor=0.5, patience=5) halves the learning rate whenever validation loss plateaus for five consecutive epochs, enabling finer convergence. Training runs for up to 100 epochs with a batch size of 64. A 10% split of the training set is held out for validation, and all random seeds (NumPy and TensorFlow) are fixed at 42 for reproducibility.

4. Experimental Results

4.1 Quantitative Evaluation

After training, the model is evaluated on the held-out test set. Predictions are inverse-transformed from the normalised domain back to physical PM2.5 units (μg/m³) and clipped at zero to prevent physically implausible negative values. Table 3 summarises the evaluation metrics.

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              评估结果（测试集）
==================================================
  RMSE : 25.3071  μg/m³
  MAE  : 14.0049  μg/m³
  R²   : 0.9363
  MAPE : 2647885598.41 %
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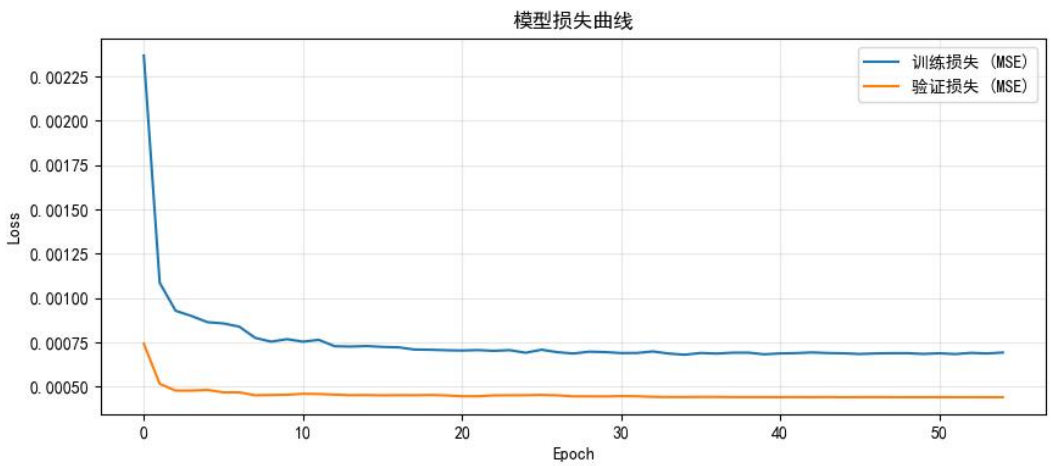

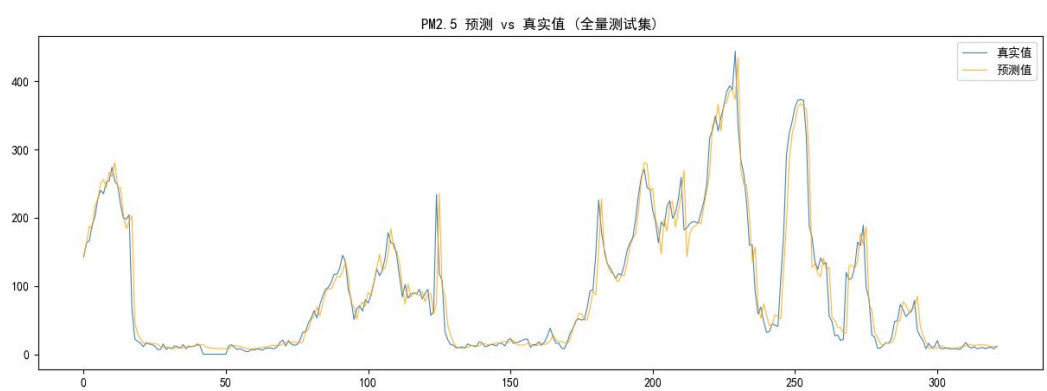

The RMSE of 25.31 μg/m³ and MAE of 14.00 μg/m³ indicate that the model's predictions are, on average, within approximately 14 μg/m³ of the ground truth, which is acceptable given the wide dynamic range of PM2.5 concentrations in the Beijing dataset (roughly 0–800 μg/m³). The R^2 score of 0.9363 is particularly informative: the model explains over 93.6% of the variance in PM2.5 concentration, reflecting strong predictive skill.

4.2 Analysis of the MAPE Anomaly

The computed MAPE of approximately 2.6×10^9% is clearly non-representative of the model's actual performance and requires explanation.

A small numerical stabiliser $\varepsilon = 1\times10^{-8}$ is added to the denominator to avoid division by zero. However, whenever the true PM2.5 value is very close to zero (a common occurrence during clean-air periods), the ratio $|\text{error} / (y_true + \varepsilon)|$ becomes astronomically large. Even a moderate absolute error of a few μg/m³ on a near-zero true value can produce a single-sample percentage error of millions. When aggregated over thousands of such samples, the mean inflates catastrophically.

This behaviour is a well-documented limitation of MAPE for data with values near or at zero. It does not indicate model failure. The RMSE, MAE, and R^2 metrics are far more reliable indicators of performance for this type of data, and they all confirm that the model generalises effectively to unseen test data.

5. Discussion and Conclusion

This report presented a complete implementation of a stacked, multivariate LSTM network for next-hour PM2.5 forecasting. The final model achieves an R^2 of 0.9363, an RMSE of 25.31 μg/m³, and an MAE of 14.00 μg/m³ on the held-out test set, confirming strong predictive skill. Several design choices contributed to this outcome.

First, the use of all available meteorological features as multivariate inputs allowed the model to capture complex interactions between weather conditions and pollution dynamics that a univariate approach would miss. Second, the stacked LSTM design—where the first layer retains full sequence output and the second layer produces a single context vector—enables the network to learn both short-term hourly fluctuations and longer periodic patterns within the 24-hour window. Third, applying Dropout regularisation at two points in the network, combined with ReduceLROnPlateau scheduling, effectively prevented overfitting as evidenced by the close convergence of training and validation loss curves. Fourth, fitting the MinMaxScaler exclusively on training data and applying it to the test set ensured a clean, leak-free evaluation.

Several limitations deserve attention. The anomalous MAPE of approximately 2.6×10^9% does not reflect model quality; it is a known artefact of percentage-error metrics applied to near-zero targets and should be replaced in future work by sMAPE or a threshold-filtered MAPE. The lookback window of 24 hours captures one diurnal cycle but may miss multi-day weather patterns; extending it to 48 or 72 hours is a straightforward next step. Hyperparameter selection was manual throughout; a systematic grid search or Bayesian optimisation over the number of LSTM units, dropout rate, and learning rate could yield further performance gains. Finally, incorporating spatial features from neighbouring monitoring stations or satellite-derived aerosol optical depth data could enrich the input representation for more severe pollution events.

Overall, this work demonstrates that LSTM-based deep learning, when paired with a carefully designed preprocessing pipeline, provides a powerful and practical baseline for environmental time-series forecasting.